Hansa Tharuka

Senior Full Stack Engineer
(Virtusa Certified AI Native Engineer)

- Senior Full Stack Engineer with **6+ years of experience** at Virtusa designing, building and supporting enterprise applications for global clients.
- **Virtusa Certified AI Native Engineer**, skilled in using AI tools across the development lifecycle to speed up coding, testing, debugging and documentation.
- Domain expertise in **Insurance** and **Human Resources (Compensation and Benchmarking)** business domains.
- Expertise in **full stack development** with Angular, Next.js, NestJS, Node.js, TypeScript, Python, Java Spring Boot and MuleSoft, backed by MongoDB, Redis and Oracle databases.
- Hands-on experience building **AI-powered applications**, including a Mastra agent service within an Nx monorepo for an AI-driven compensation market-pricing platform.
- Strong background in **microservices, API gateways and secure integration**: OAuth 2.0, JWT, rate limiting, TLS/SSL and AES-256 encryption, with OWASP and GDPR compliance.
- Experienced in **platform migrations and secrets management**: IBM WebSphere to WildFly, Apache Kafka to AWS MSK, and HashiCorp Vault to CyberArk Conjur.
- Proficient in **cloud and DevOps** (AWS EKS, MSK and EC2, Docker, Kubernetes, Jenkins, GitLab CI, Bamboo) and in production monitoring with Datadog, Komodor, Sumo Logic, New Relic, Grafana and Kibana.
- Quick to adapt to new technologies and methodologies, working in **Agile/Scrum** teams with a focus on clean code, design patterns and stable, scalable releases.
- Communicates effectively with clients, external partners and all levels of the organization, from senior leadership to team members.

Core Competencies

- Full stack web application design and development
- Microservices and API design (REST, SOAP)
- AI-native engineering and AI-assisted development
- Enterprise integration and API gateways with MuleSoft
- Production support, root-cause analysis and incident resolution
- Application, platform and messaging migrations
- AI agent services and AI-driven analytics (Mastra)
- Secure configuration and secrets management
- CI/CD, containerisation and cloud deployment
- Testing and quality: unit, end-to-end and load testing
- Impact and risk analysis, documentation and knowledge transfer
- Agile/Scrum delivery and client communication

Tools and Technologies

- **Angular, Next.js, TypeScript, HTML5, CSS3** – For frontend web application development
- **NestJS, Node.js, Python, Java, Spring Boot, Spring Cloud** – For backend services and microservices
- **Mastra, Databricks** – For AI agent services and data processing

- **MuleSoft (Anypoint Studio, API Manager, Runtime Fabric), DataWeave** – For API gateways and enterprise integration
- **MongoDB, Redis, Oracle DB, MySQL, Toad** – For data storage, caching and querying
- **AWS (EKS, MSK, EC2), Docker, Kubernetes, Apache Kafka** – For cloud deployment, containerisation and messaging
- **Jenkins, GitLab CI, Bamboo, Nx, Maven, Gradle, Git, SVN** – For CI/CD, builds, monorepo management and version control
- **Jest, Cypress, JUnit, Selenium, ReadyAPI, SoapUI, Postman** – For unit, end-to-end and API testing
- **Datadog, Komodor, Sumo Logic, New Relic, Grafana, Kibana, ELK Stack, Splunk** – For monitoring, logging and alerting
- **HashiCorp Vault, Consul, CyberArk Conjur, Wiz** – For secrets, configuration management and security scanning
- **WildFly/JBoss, Tomcat, IBM WebSphere** – Application servers for deployment and migration
- **JIRA, PractiTest, Swagger** – For Agile project tracking, test tracking and API documentation

Project Responsibilities at Virtusa

- Design, develop and enhance full stack features across Angular and Next.js frontends and NestJS, Spring Boot and MuleSoft backends.
- Build and maintain AI-enabled services, including a Mastra agent service supporting automated benchmarking, peer-group resolution and regression-based pay recommendations.
- Analyse production issues reported by business support teams, identify affected components, perform impact and risk analysis, and deliver fixes through lower environments, QA and production release with post-production validation.
- Design and implement secure APIs and gateways with authentication, authorization, traffic management and data encryption.
- Plan and execute platform, messaging and secrets-management migrations with impact analysis and risk assessments to minimise downtime.
- Manage deployments across environments and monitor application performance using Datadog, Komodor, Sumo Logic and New Relic, resolving bottlenecks and production issues.
- Write and maintain unit, end-to-end and load tests, address code-review feedback, update Swagger documentation and resolve security scan findings.
- Work with clients, external partners and cross-functional teams (e.g. New York Life, Fineos, Global Business Services) on technical support, debugging and issue resolution.

Organizational Responsibilities

- Conduct knowledge-transfer sessions and prepare handover documentation for migrated and newly built services.
- Provide technical guidance and debugging support to team members and partner teams.
- Contribute to engineering best practices, including clean code, design patterns and code quality.

Projects

Virtusa Compensation Assistant – AI

Role	Senior Software Engineer (Full Stack) | Business / Human Resources
Technology	Angular 20, Next.js, NestJS, Node.js, TypeScript, Python, Mastra, MongoDB, Redis, Databricks, Nx, Jest, Cypress, Datadog, Komodor
Project Description	AI-powered market-pricing platform for compensation analysis, built as an Nx monorepo with an Angular 20 frontend, NestJS/MongoDB API and a Mastra agent service. Automates benchmarking, peer-group resolution and regression-based pay recommendations.
Responsibilities	• **Benchmark Export and Reporting:** Developed and fixed export features for benchmark charts and regression results; updated export endpoints, file naming and pay-component values, and fixed the missing negative sign for below-market figures. • **Defect Resolution and UI Refinement:** Fixed defects in the default benchmarking stage, including title casing, data card expansion and pay-component ordering. • **Code Quality, Testing and Security:** Addressed code-review feedback, wrote and repaired Jest unit tests, updated Swagger documentation, fixed build issues and upgraded dd-trace to clear a Wiz security finding. • **Deployment and Monitoring:** Managed deployments across environments and monitored performance with Datadog and Komodor, fixing bottlenecks and production issues.

Cross Tech

Role	Senior Software Engineer | Insurance
Technology	Java, Spring Boot, MuleSoft, MuleSoft Runtime Fabric, CyberArk Conjur, Sumo Logic, AWS
Project Description	Identified and resolved production issues in MuleSoft and Spring Boot applications, and built urgent new components and projects from scratch to meet business needs.
Responsibilities	• **Production Issue Management:** Resolved issues reported by business support end to end: root-cause and risk analysis, fixes in lower environments, QA handover, production release and post-release validation. • **New Component Development:** Designed, built and tested new components and projects from scratch through to production release and support. • **Risk Analysis and Communication:** Ran impact and risk analysis for fixes and new work, keeping stakeholders and cross-functional teams informed. • **End-to-End Delivery with QA:** Coordinated QA testing, released to production and monitored post-release performance.

Vault to Conjur Migration

Role	Senior Software Engineer | Insurance
Technology	Java, Spring Boot, MuleSoft, MuleSoft Runtime Fabric, CyberArk Conjur, HashiCorp Vault, WildFly, Sumo Logic, AWS, Grafana, Kibana

Project Description	Planned and executed the migration of secure properties from HashiCorp Vault to CyberArk Conjur for Spring Boot and MuleSoft services.
Responsibilities	• **Configuration Service Update:** Extended the configuration service to integrate with CyberArk Conjur for secure secret management. • **Reusable Java Libraries:** Built Java dependencies for Spring Boot applications, including a WildFly-specific variant, to load configuration at startup and reduce startup time. • **Migration Plan:** Wrote the migration plan with impact analysis and risk assessment, coordinating teams to minimise downtime. • **Monitoring Dashboards:** Built Grafana and Kibana dashboards to track migrated services. • **Handover and Documentation:** Handed over services with documentation and knowledge-transfer sessions.

External Gateway

Role	Senior Software Engineer | Insurance
Technology	MuleSoft, MuleSoft Runtime Fabric, Java, Shell Script, Sumo Logic, Consul, HashiCorp Vault, AWS EKS
Project Description	Developed an API gateway with MuleSoft and Java for secure, efficient communication and data integration between systems.
Responsibilities	• **Secure API Design:** Built secure APIs with OAuth 2.0, JWT and API-key authentication. • **API Traffic Management:** Configured rate limiting, throttling and traffic shaping with MuleSoft API Manager. • **Data Encryption:** Applied TLS/SSL in transit and AES-256 at rest. • **Monitoring and Logging:** Set up ELK Stack, Splunk and Anypoint Monitoring with real-time alerts. • **Compliance:** Ensured OWASP and GDPR compliance through regular security audits and vulnerability assessments.

Foundation

Role	Software Engineer | Insurance
Technology	MuleSoft, Java Spring Boot, Spring Cloud, Bash Script, AWS (EKS, MSK, EC2), Apache Kafka, Kerberos, Sumo Logic, New Relic, Consul, HashiCorp Vault, Jenkins, GitLab CI, Bamboo
Project Description	Foundational platform work covering application management, CI/CD configuration, migrations, gateway applications and support for clients and external partners.
Responsibilities	• **Gateway Proxies and Change Requests:** Managed MuleSoft gateway proxies and delivered API integration change requests with thorough deployment testing. • **CI/CD Pipelines:** Maintained Jenkins, GitLab CI and Bamboo jobs and scripts, reducing deployment time and errors. • **Technical Support and Log Analysis:** Supported external partners (Fineos), internal teams (Global Business Services) and the New York Life team with debugging and detailed log checks. • **Spring Boot Development:** Implemented enhancements in Spring Boot microservices using Spring Cloud.

	• **Kafka to AWS MSK Migration:** Migrated the communication API from Apache Kafka to Amazon MSK, improving throughput and reliability. • **Configuration Management:** Managed application configuration with Consul and HashiCorp Vault.

WAS Migration

Role	Software Engineer | Insurance
Technology	Java, Oracle DB, IBM WebSphere, WildFly, Consul, HashiCorp Vault, Sumo Logic, New Relic, JUnit, PractiTest
Project Description	Migrated legacy services from IBM WebSphere to WildFly and upgraded Java from version 6 to 8, fixing defects throughout.
Responsibilities	• **Legacy Analysis and Migration:** Analysed legacy code to plan the WebSphere-to-WildFly move and Java upgrade. • **POC Development:** Built proofs of concept integrating Consul and Vault for configuration at startup. • **Testing:** Fixed post-migration bugs and ran functional, JUnit and load tests to remove performance bottlenecks. • **Database and Payload Testing:** Queried Oracle databases to test payloads and fix data issues. • **Monitoring and Client Reporting:** Debugged issues with Sumo Logic, tracked work in PractiTest and reported fixes to clients.

Educational Qualifications

- BSc (Hons) in Information Technology – University of Moratuwa, Sri Lanka (GPA 3.01)
- International Diploma in Information Technology – Esoft Metro Campus (Pearson UK / Edexcel)

Achievements & Certifications

- Virtusa Certified AI Native Engineer | Virtusa